CLINICAL · NOSE & SINUSES

# Granulomatous disease of the nose

A crusted, destroyed nose that will not heal has a long differential: chronic infections such as rhinoscleroma, tuberculosis, leprosy, syphilis and rhinosporidiosis; inflammatory disease such as granulomatosis with polyangiitis and sarcoidosis; and NK/T-cell lymphoma, which looks like inflammation but is a cancer. Many are systemic diseases showing in the nose, and none can be named without a biopsy.

AT A GLANCE

- Crusting and destruction need a tissue diagnosis.
- Check kidneys, lungs and ANCA.
- Exclude lymphoma, infection and cocaine.
- Treat disease before repairing the nose.

## 1 CHAPTER ONE Overview

The destructive, crusting nose.

### 1.1 Classification

GRANULOMATOUS DISEASES OF THE NOSE

- BACTERIAL **Rhinoscleroma** tuberculosis, lupus vulgaris, leprosy, syphilis
- FUNGAL **Aspergillosis, mucormycosis** see the fungal sinusitis topic
- PARASITIC **Rhinosporidiosis** leishmaniasis
- INFLAMMATORY **Granulomatosis with polyangiitis** sarcoidosis, eosinophilic granulomatosis
- NEOPLASTIC MIMIC **NK/T-cell lymphoma** the old lethal midline granuloma

### 1.2 The rule

Always biopsy, and always look for the systemic disease: chest, kidneys, eyes, skin and lymph nodes. Many of these conditions mimic a neoplasm, and a neoplasm can mimic them.

### 1.3 Common findings

Crusting, bloodstained discharge, nasal obstruction, granulations, septal perforation, saddle nose and, in some, palatal destruction.

## 2 CHAPTER TWO Bacterial infections

Chronic and destructive.

### 2.1 Rhinoscleroma

#### 2.1.1 Cause

*Klebsiella rhinoscleromatis* (Frisch bacillus), a Gram-negative bacillus; endemic in parts of Africa, Central America, Eastern Europe, the Middle East (including Egypt) and North India.

#### 2.1.2 Spread

Starts in the nose and spreads to the nasopharynx, oropharynx, larynx (especially the subglottis), trachea and bronchi.

#### 2.1.3 Stages

- **Atrophic (catarrhal):** like atrophic rhinitis, with foul purulent discharge and crusts.
- **Granulomatous (proliferative):** painless, non-ulcerating nodules; infiltration of the lower external nose and upper lip gives a *woody* feel (the *Hebra nose*).
- **Cicatricial (sclerotic):** stenosis of the nostrils, distortion of the lip, adhesions in the nose and pharynx, and subglottic stenosis with stridor.

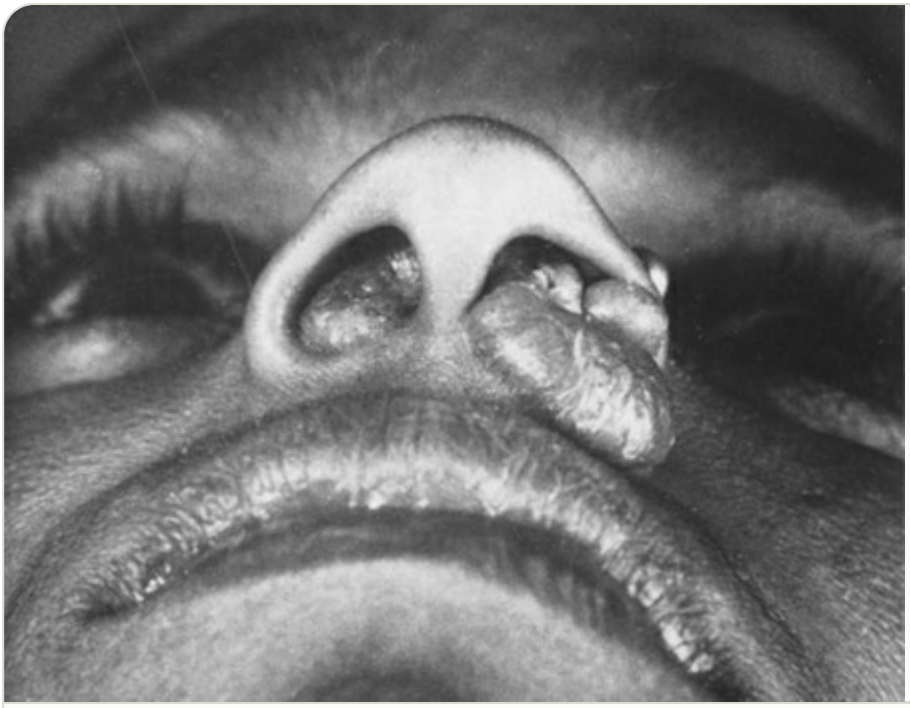


**FIGURE 1** Rhinoscleroma of the nose.
Dhingra, fig. 28.1

### 2.1.4 Diagnosis

Biopsy: submucosal plasma cells, lymphocytes and eosinophils, with *Mikulicz cells* (large foamy macrophages containing the bacilli) and *Russell bodies* (eosinophilic inclusions of immunoglobulin in plasma cells), which are diagnostic. Culture of the biopsy.

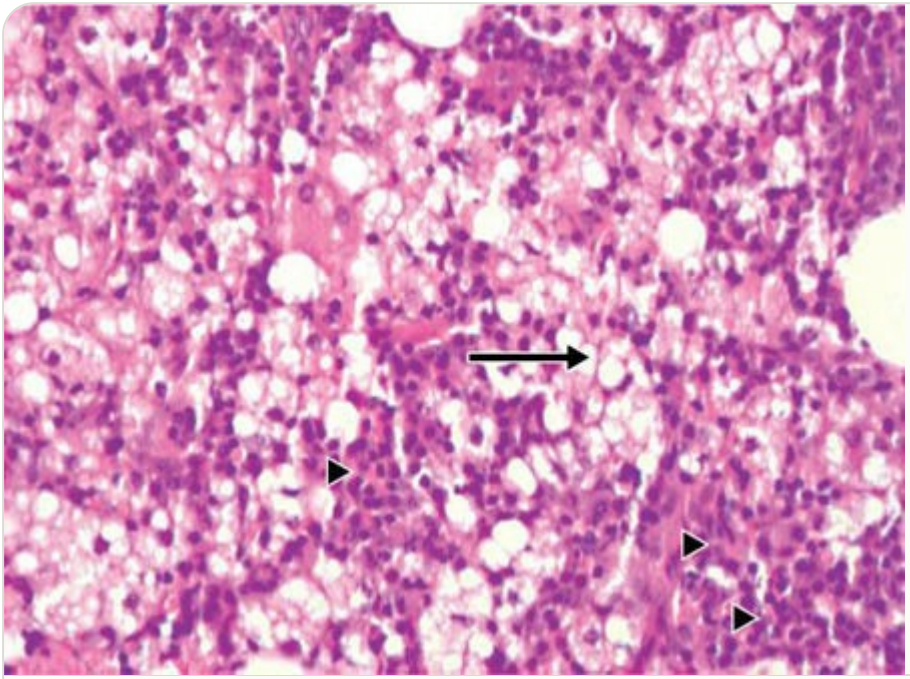


**FIGURE 2** Rhinoscleroma: foamy Mikulicz cells (arrow) among lymphocytes (arrowheads).
Dhingra, fig. 28.2

#### 2.1.5 Treatment

Long antibiotic courses until two consecutive biopsy cultures are negative.

WHERE THE BOOKS DIFFER · ANTIBIOTICS FOR RHINOSCLEROMA

| Dhingra | Lee |
| --- | --- |
| Streptomycin 1 g/day with tetracycline 2 g/day for at least four to six weeks, repeated if needed; steroids to reduce fibrosis. | Fluoroquinolones such as ciprofloxacin, or tetracyclines, for months. |

Surgery restores the airway (including tracheostomy for subglottic stenosis) and corrects deformity.

## 2.2 Tuberculosis

Usually secondary to pulmonary disease. Nodular infiltration, then ulceration of the anterior septum and anterior end of the inferior turbinate, with perforation of the cartilaginous septum. Biopsy with acid-fast stains and culture; antituberculous drugs.

## 2.3 Lupus vulgaris

A low-grade tuberculous infection of the vestibule and the skin of the nose and face, with brownish *apple-jelly nodules* and chronic vestibulitis; may perforate the cartilaginous septum. Biopsy; antituberculous treatment.

## 2.4 Leprosy

*Mycobacterium leprae*, mainly in the lepromatous form. The anterior septum and inferior turbinate become red and swollen with excess discharge, then crusting, bleeding and nodules that ulcerate and perforate. Late sequelae: atrophic rhinitis, a depressed bridge and loss of the anterior nasal spine with a retracted columella. Diagnosis from scrapings and biopsy (lepra bacilli in foamy *lepra cells*); multidrug therapy with dapsone and rifampicin; reconstruction when inactive.

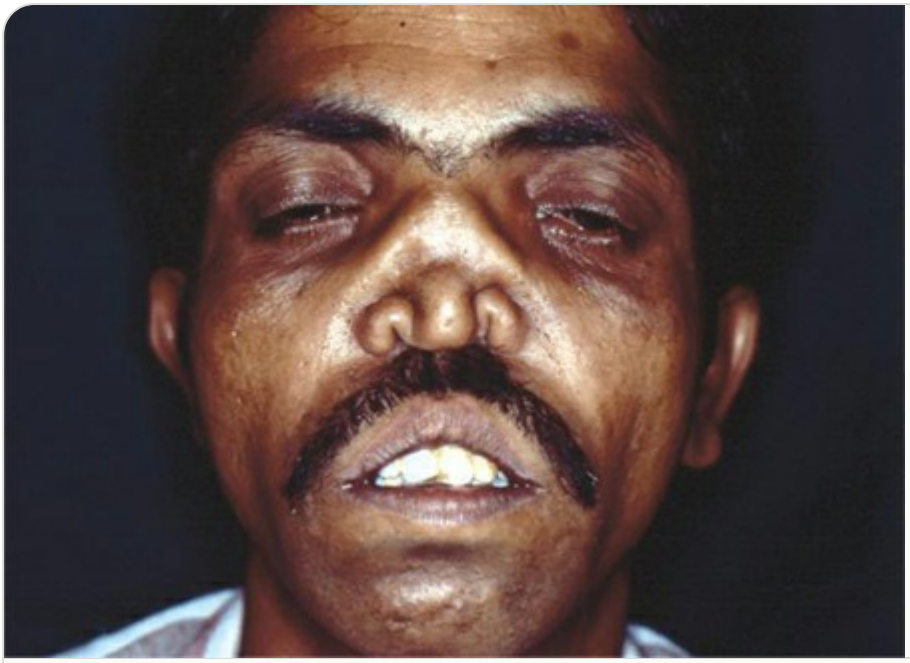


**FIGURE 3** Leprosy of the nose.
Dhingra, fig. 28.3

## 2.5 Syphilis

### 2.5.1 Acquired

- **Primary:** a rare chancre of the vestibule.
- **Secondary:** rhinitis with crusting and fissuring, with mucous patches, rash and lymphadenopathy.
- **Tertiary:** the stage that involves the nose: a *gumma* destroys the septum, both bony and cartilaginous, with sequestra, foul crusts, palatal perforation and a saddle nose.

### 2.5.2 Congenital

- **Early:** *snuffles* in the first three months, becoming purulent, with excoriation of the vestibule and lip.
- **Late:** around puberty, with gummatous destruction like tertiary disease, plus corneal opacities, deafness and Hutchinson's teeth.

### 2.5.3 Diagnosis and treatment

Serology (VDRL with treponemal tests) and biopsy. Benzathine penicillin 2.4 million units weekly for three weeks; alkaline douches; removal of sequestra; reconstruction once inactive.

**WHICH PART OF THE SEPTUM?**

Tuberculosis, lupus and leprosy perforate the cartilaginous septum; syphilis destroys the bony septum.

# 3 CHAPTER THREE Parasitic infections

Rhinosporidiosis and leishmaniasis.

## 3.1 Rhinosporidiosis

### 3.1.1 Cause and spread

*Rhinosporidium seeberi*, once thought a fungus, now classed as an aquatic protistan parasite. Acquired from ponds shared with animals; common in South India and Sri Lanka.

### 3.1.2 Clinical picture

A leafy, polypoidal, pink to purple, strawberry-like mass on the septum or lateral wall, very vascular and bleeding easily, its surface studded with white dots (the sporangia). Bloodstained discharge or epistaxis may be the only complaint; it may hang into the nasopharynx.

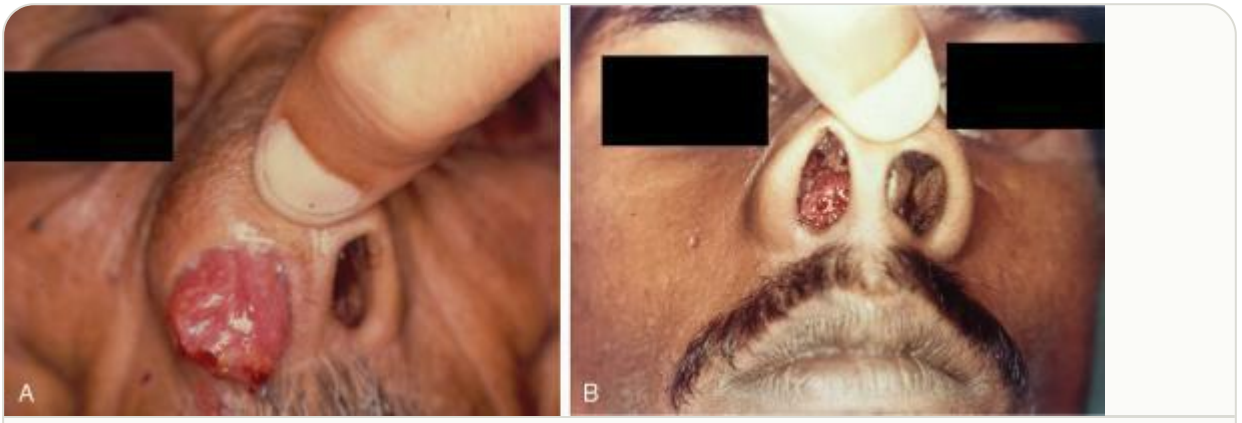



**FIGURE 4** Rhinosporidiosis: a polypoidal mass protruding from the nostril, and its appearance inside the nose.
Dhingra, fig. 28.6

### 3.1.3 Life cycle

Endospores grow into trophocytes, which mature into *sporangia* 200 to 250 µm across containing thousands of endospores; the sporangium bursts through a pore and releases them.

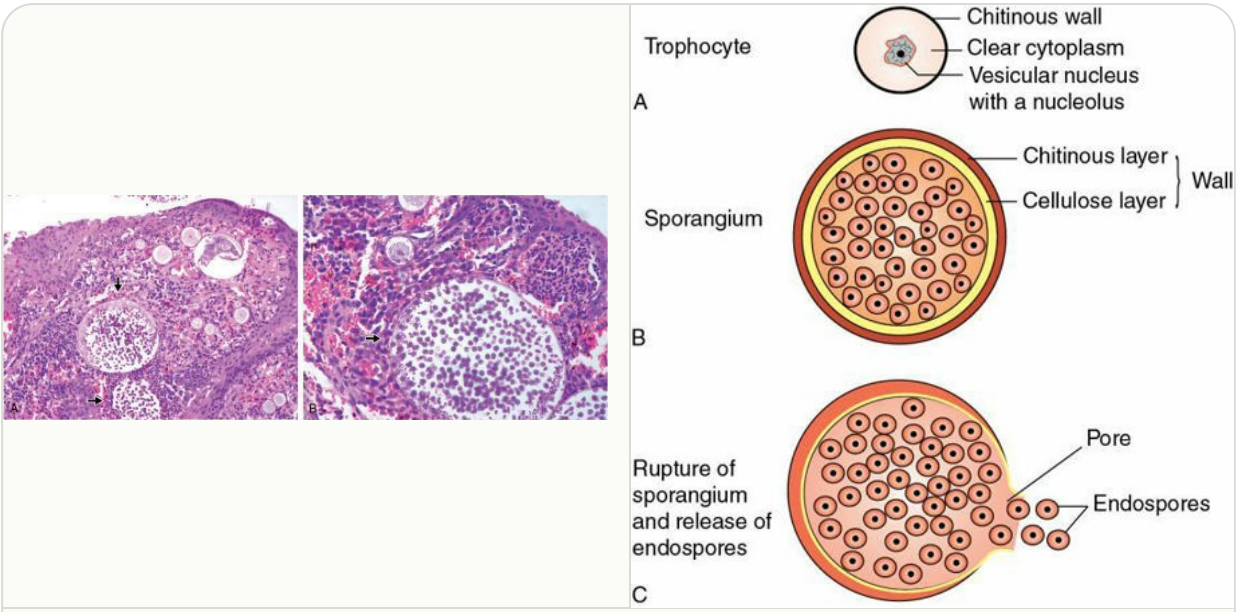



**FIGURE 5** First, histology with sporangia packed with spores. Second, the life cycle of the organism.

Dhingra, figs 28.7 and 28.8

### 3.1.4 Diagnosis and treatment

Biopsy shows the sporangia; the organism cannot be cultured. Complete excision with a diathermy knife and cautery of the base; recurrence is common; dapsone helps some patients.

## 3.2 Leishmaniasis

Transmitted by the sandfly (*Phlebotomus*). Cutaneous lesions on exposed skin progress from papules to crusted ulcers; mucosal disease appears on the septum as a nodule or plaque that ulcerates and perforates, sometimes years after a skin lesion, and can spread to the palate, pharynx and larynx. Biopsy, culture and PCR; mucosal disease needs parenteral antimonials (sodium stibogluconate or meglumine antimoniate).

# 4 CHAPTER FOUR Inflammatory diseases

Vasculitis and sarcoid.

## 4.1 Granulomatosis with polyangiitis

### 4.1.1 What it is

Formerly Wegener's granulomatosis: a necrotising granulomatous vasculitis of small vessels affecting the upper airway, lungs and kidneys (the classic triad), with skin, eye and ear involvement.

### 4.1.2 Nose and sinuses

A persistent cold or sinusitis, bloodstained then purulent discharge, crusting, granulations, septal perforation and saddle nose; destruction may extend to the orbit, palate and middle ear. Subglottic stenosis develops in some.

### 4.1.3 Systemic features

Fatigue, night sweats, arthralgia, anaemia; cough and haemoptysis with cavitating lung nodules; glomerulonephritis with blood, casts and protein in the urine and a rising creatinine, the usual cause of death untreated.

### 4.1.4 Diagnosis

*c-ANCA* (anti-proteinase 3) is positive in most active systemic disease; raised ESR and CRP; urinalysis and creatinine; chest imaging. Nasal biopsy shows necrosis, granulomas and vasculitis, though often non-specific.

### 4.1.5 Treatment

Steroids with cyclophosphamide or rituximab to induce remission, then azathioprine or methotrexate to maintain it, with the rheumatology and renal teams. Nasal douching for crusts; reconstruction only when quiescent.

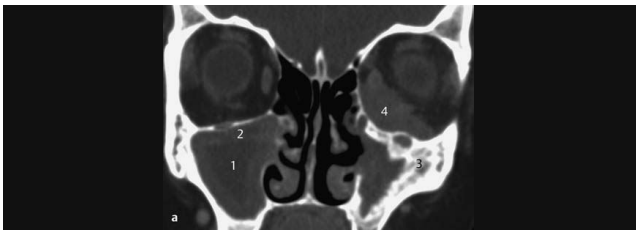
**IMAGING IN THE APP** 2 cases

Granulomatosis with polyangiitis in the nose, sinuses and orbit.




**Granulomatosis with Polyangiitis (Wegener)**

Infection & inflammation · 2 images




**Granulomatosis with Polyangiitis, Orbital Involvement**

Infection & inflammation · 2 images

## 4.2 Eosinophilic granulomatosis with polyangiitis

Churg-Strauss syndrome: asthma, nasal polyps, peripheral eosinophilia and vasculitis, often p-ANCA positive.

## 4.3 Sarcoidosis

A systemic non-caseating granulomatous disease. In the nose: submucosal nodules on the septum and inferior turbinate with obstruction, crusting, pain and bleeding; nodules on the vestibule and facial skin (*lupus pernio*). Chest imaging shows hilar adenopathy; serum calcium and *serum ACE* are raised in active disease. Topical and systemic steroids.

# 5 CHAPTER FIVE NK/T-cell lymphoma

The cancer that looks like inflammation.

## 5.1 What it is

Extranodal NK/T-cell lymphoma, nasal type, associated with Epstein-Barr virus; the condition once called *lethal midline granuloma* or polymorphic reticulosis. Commoner in East Asia and Latin America.

## 5.2 Clinical picture

Progressive, relentless destruction of the midline: nasal obstruction and discharge, ulceration and necrosis of the septum, palate and facial skin, with fever and weight loss.

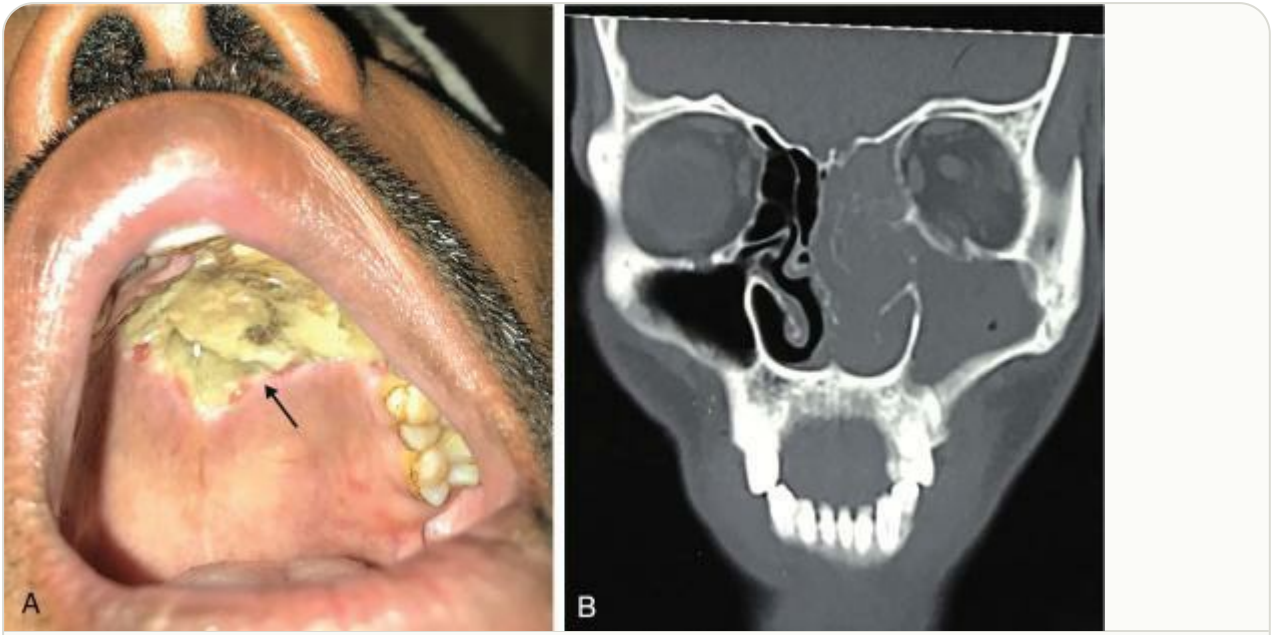



**FIGURE 6** NK-cell lymphoma with midline destruction in a young man.
Dhingra, fig. 28.9

## 5.3 Diagnosis

Deep, repeated biopsies of viable tissue, with immunohistochemistry (CD56, cytotoxic markers) and EBV studies; superficial biopsies show only necrosis and inflammation.

## 5.4 Treatment

Radiotherapy with chemotherapy. It must be told apart from granulomatosis with polyangiitis, because the treatments are completely different.

# 6 CHAPTER SIX Assessment

The work-up of a destructive nose.

### INVESTIGATIONS BY SUSPICION

| SUSPECTED DISEASE | KEY TEST |
|---|---|
| Granulomatosis with polyangiitis | c-ANCA, urinalysis, creatinine, chest imaging, biopsy |
| Sarcoidosis | Serum ACE, calcium, chest imaging |
| Tuberculosis | Acid-fast stains, culture, PCR, chest imaging |
| Syphilis | VDRL with treponemal tests |
| Rhinoscleroma | Biopsy for Mikulicz cells and Russell bodies, culture |
| Lymphoma | Deep biopsy with immunohistochemistry and EBV studies |
| Cocaine damage | History and urine toxicology |

**COCAINE**

Cocaine-induced midline destruction can mimic vasculitis, and some patients develop ANCA from levamisole in the drug. Ask, and test.

# 7 CHAPTER SEVEN For the examination

What the examiner wants to hear.

## 7.1 Viva essentials

**WHAT THE EXAMINER IS LOOKING FOR**

- **Classification** of granulomatous disease of the nose.
- **Rhinoscleroma:** organism, stages, Mikulicz cells and Russell bodies, treatment.
- **Rhinosporidiosis:** organism, appearance, sporangia, treatment.
- **Syphilis of the nose:** stages, snuffles, saddle nose, bony septal perforation.

- **Granulomatosis with polyangiitis:** triad, c-ANCA, treatment.
- **NK/T-cell lymphoma** versus vasculitis.

## 7.2 Common pitfalls

COMMON PITFALLS

- Treating a destructive nose with antibiotics for months without a biopsy.
- Superficial biopsies of necrotic tissue in suspected lymphoma.
- Forgetting the kidneys in granulomatosis with polyangiitis.

Written for this app from Oxford Textbook of Otolaryngology ch. 34, Oxford Specialist Handbook ch. 20, Oxford Handbook of ENT ch. 9, Dhingra ch. 28 and K. J. Lee ch. 12. Figures reproduced from the books for private study. Study reference only: follow local protocols and senior advice for patients.

IMAGING

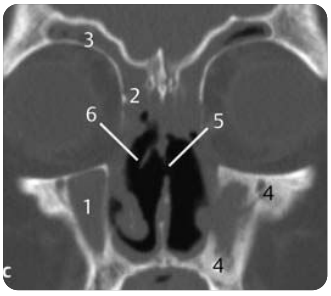
# On imaging

*The imaging of this topic: key findings, differentials and representative scans.*

NOSE & NASAL CAVITY · INFECTION & INFLAMMATION

## Granulomatosis with Polyangiitis (Wegener)

- A history of chronic sinusitis in combination with destruction of the nasal septum is highly suggestive of granulomatosis with polyangiitis (GPA). Pulmonary complaints and rapid improvement after steroid treatment may confirm this diagnosis. These therapeutic effects may also be observed in cases of allergic fungal sinusitis.
- Orbital involvement appears as a mass resulting in protrusion of the eyeball and diplopia (see also granulomatosis with polyangiitis (GPA), Orbital Involvement).

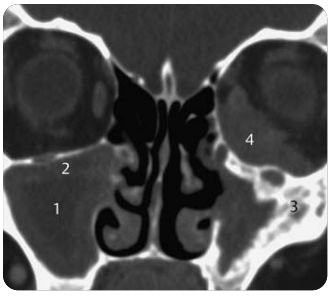



Patient with sinus pressure, nasal discharge, crusting, hyposomia, and minor epistaxis.

Dunnebier, Imaging for…

NOSE & NASAL CAVITY · INFECTION & INFLAMMATION

## Granulomatosis with Polyangiitis, Orbital Involvement




Patient with known granulomatosis with polyangiitis (GPA) and progressive restriction of ocular movement….